PACLAND · Economics

The verification discipline applied to macro-finance — a plain-English brief

The headlines — and what to do about each

‣ 6 named models are scored on a nine-slot bar; 4 are clean (random walk, Godley–Lavoie SFC, VAR(1), GDPNow). Scoring is mechanical: a claim with no evidence behind it is refused, not argued with.

‣ On real-time data the no-change null beats structural inflation forecasts: a calibrated three-equation model lands at 1.03× the null's error, and the same model at Smets–Wouters' published parameters does no better. A plain VAR beats the null (0.86×), and the Atlanta Fed's published nowcast beats it 0.25× on the advance estimate, 0.31× after revisions.

‣ The Phillips slope cannot be identified from aggregate data: the estimator cannot recover a known slope of 0.18 from the model's own simulated data (it returns 1.00). That is not a data-quantity problem — more data will not fix it.

‣ The exact content is definitional, and that is the finding: the nominal expenditure identity closes to ~1e-4% of GDP, while the *real* decomposition leaks by up to 25 percentage points in 2020 because chained dollars are not additive.

‣ 11 constraints, 8 independent — and they refuse a data-fitted passive policy rule outright instead of letting it into a forecast table.

‣ Two of this discipline's own claims were retracted on contact with evidence: that the workhorse DSGE cannot beat a VAR out of sample (its own Table 3 disproves it), and that GDPNow's edge is an artifact of targeting the first print (it survives revisions). Both retractions are in the record.

1 · What we are measuring

Macroeconomics forecasts a future that is never observed, with data that is revised after the fact. The discipline asks the question the field does not put in numbers: how much of a forecast is verified, and how much is extrapolation?

2 · The three things we built

- A bar with nine slots — exact structure, a deficit count, three independent checks, a mutation-tested guard-suite, an out-of-sample run, a benchmark, a board tier, a falsifiable prediction, and one reproduce command. A model's kind decides which it owes.
- A null. 'Tomorrow equals today.' Every model is scored against it on real-time vintages — what was published at the time, never today's revised numbers.

- A constraint set: 11 constraints of which 8 are independent, measured by the rank of the constraint Jacobian.

3 · The verification deficit

- 4 of 10 backcast series are measured + computable + validated — a certificate radius of 68%.
- 0 of 5 forecast targets are validated — the projection central banks act on is extrapolation, not verification.

A 1% certificate needs ~18,000 validated observables. The bottleneck is measured data, not compute, and no model in the roster can escape that.

4 · Why the null matters more than the model

The field's own record is the anchor: Meese–Rogoff (1983) found structural exchange-rate models fail to beat a random walk out of sample, and Rossi (2013) confirms the pessimism. What this work adds is the measurement — on real-time data, for named models, with the null in the same table. Some models do beat it (a VAR, and the published nowcast by a wide margin); the structural forecasts of inflation do not.

5 · What it means, and what is possible

The product is not a forecast; it is a set of gates. A null gate (no forecast ships without its no-change ratio, and losing it is stated in the tier). An identification gate (no estimate ships unless the estimator can recover a known truth from synthetic data). An admissibility gate (no fitted model enters a table if it violates the constraint set). All three have already refused something.

What is possible: a verified-forecast label any macro model can carry — its m observables, its constraint pass/fail, and its real-time ratio to the null. Nobody publishes that today. What is not possible, on this evidence, is a certified quarter-ahead inflation number: the exact content is definitional and the behavioural parameters are not identified from aggregates.

6 · Reproduce

pacland economics --models · --constraints · --realtime · docs/economics-model-roster.md